EXPERIMENT: SETUP AND CONFIGURATION OF A LAN USING A SWITCH AND ETHERNET CABLES

AIM

To set up and configure a Local Area Network (LAN) using a switch and Ethernet cables in the laboratory.

WHAT IS A LAN?

A Local Area Network (LAN) connects devices within a limited area, such as an office, school, or home. It enables users to share resources, including data, printers, and internet access. Devices such as computers, printers, servers, and switches are connected to facilitate communication and information transfer. A LAN switch acts as the primary connecting device and manages communication between connected devices, enabling fast data transfer.

REQUIREMENTS

- Four computers
- One network switch with 8, 16, or 24 ports
- Four Ethernet cables
- Network interface cards
- Windows operating system

PROCEDURE

Step 1: Plan the Network Topology

Plan and design a suitable network topology based on the network requirements and equipment location.

Step 2: Arrange the Devices

Take four computers, one network switch, and four Ethernet cables.

Step 3: Connect the Computers

Connect each computer to the network switch using an Ethernet cable. Plug one end of the cable into the computer's Ethernet port and the other end into the switch.

Step 4: Assign IP Addresses to the PCs

1. Log in to each client computer as Administrator or Owner.
2. Open Network and Internet Connections.
3. Right-click Local Area Connection/Ethernet and select Properties.
4. Select Internet Protocol (TCP/IPv4) and click Properties.
5. Select Use the following IP address.
6. Assign the following IP addresses and subnet masks to the computers.

Device	IP Address	Subnet Mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0

Step 5: Configure the Network Switch

1. Connect a computer to the switch using an Ethernet cable.
2. Open a web browser and enter the switch's IP address in the address bar.
3. Log in to the web interface using the required username and password.
4. Configure the basic settings of the switch.
5. Assign the IP address 10.1.1.5 with subnet mask 255.0.0.0 to the switch management interface.

Step 6: Check Connectivity

Open the Command Prompt on a computer and use the `ping` command to check connectivity between the computers and the switch management IP address.

Example:

```
ping 10.1.1.2
ping 10.1.1.3
ping 10.1.1.4
ping 10.1.1.5
```

Step 7: Share a Folder

1. Select a folder on a computer.
2. Right-click the folder and select Properties.

3. Open the Sharing tab.
4. Share the folder with everyone on the same LAN.

Step 8: Access the Shared Folder

From another computer on the network, access the shared folder and verify that file sharing works correctly.

NETWORK CONFIGURATION

Device	IP Address	Subnet Mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0
Switch	10.1.1.5	255.0.0.0

SCREENSHOTS:

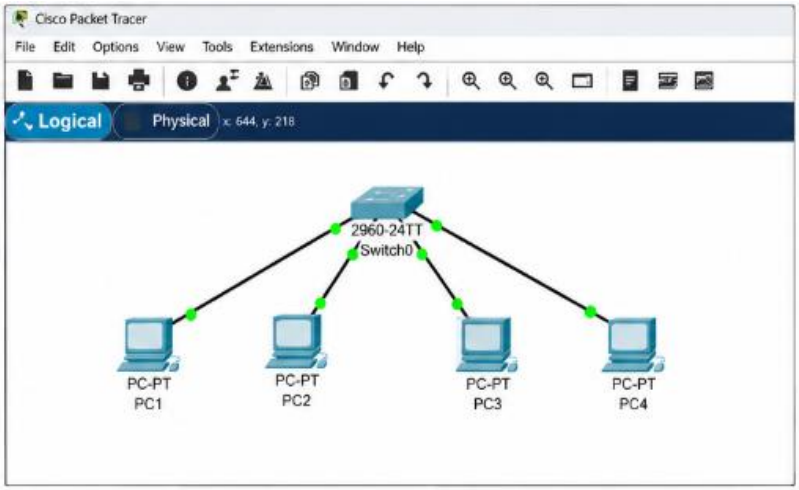


Fig 1: LAN Topology with 4 PCs and a Switch

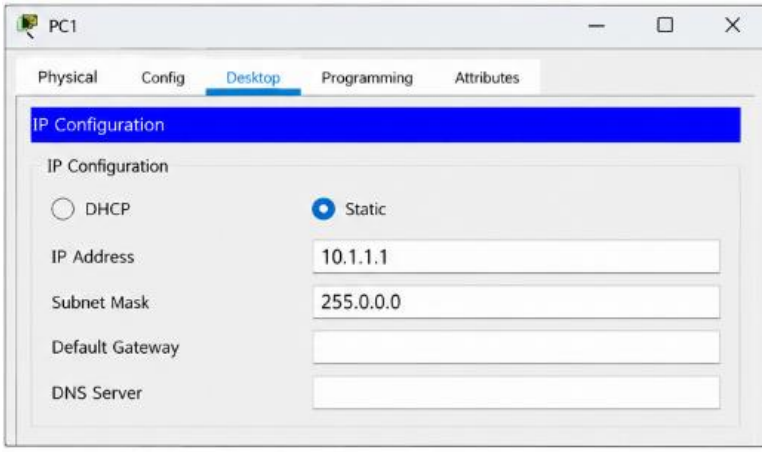


Fig 2: IP Configuration of PC1 (10.1.1.1)

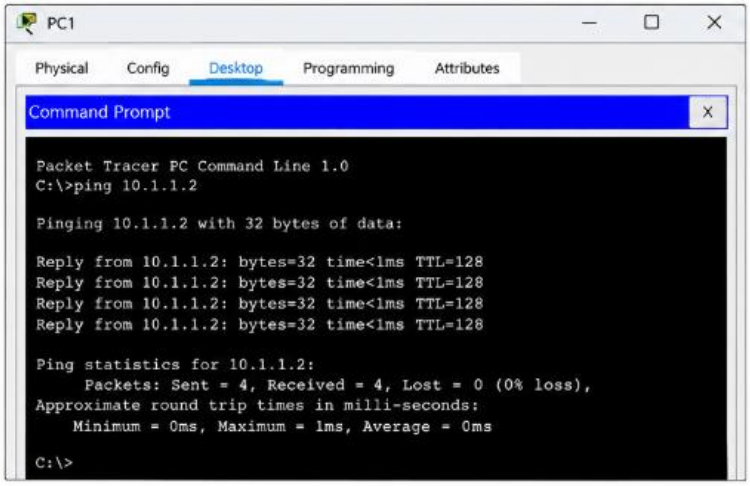


Fig 3: Ping Output (PC1 to PC2)

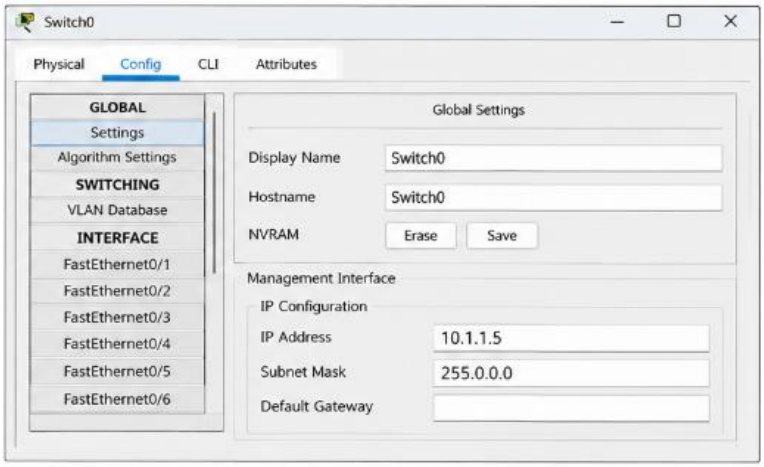


Fig 4: Switch Management IP Configuration (10.1.1.5)

RESULT

A Local Area Network was set up and configured using a network switch and Ethernet cables. IP addresses were assigned to the connected devices, network connectivity was checked using the ping command, and folder sharing was tested between computers.